Computer Science & IT



C Programming

Control Flow Statement

Lecture No. 01



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Recap of Previous Lecture



Topic Increment & Decrement Operator

Topic Bitwise Operator

Topic

Topic

Topic

Slide

Topics to be Covered



- Topic — Printf function
- Topic — Ternary function
- Topic — Control Statements
- Topic
- Topic

Slide

bit wise Complement operator

int x = 5

0 0 0 0 0 1 0 1

a = ~x

0 0 0 0 0 1 0 1

| | | | | | | |

1 1 1 1 1 0 1 0 (MSB ..) Negative

hhu

1 Byte

unsigned

11111010 ← Negative 1's

↓ 2's complement

1 Byte

11111010 (−6) ✓

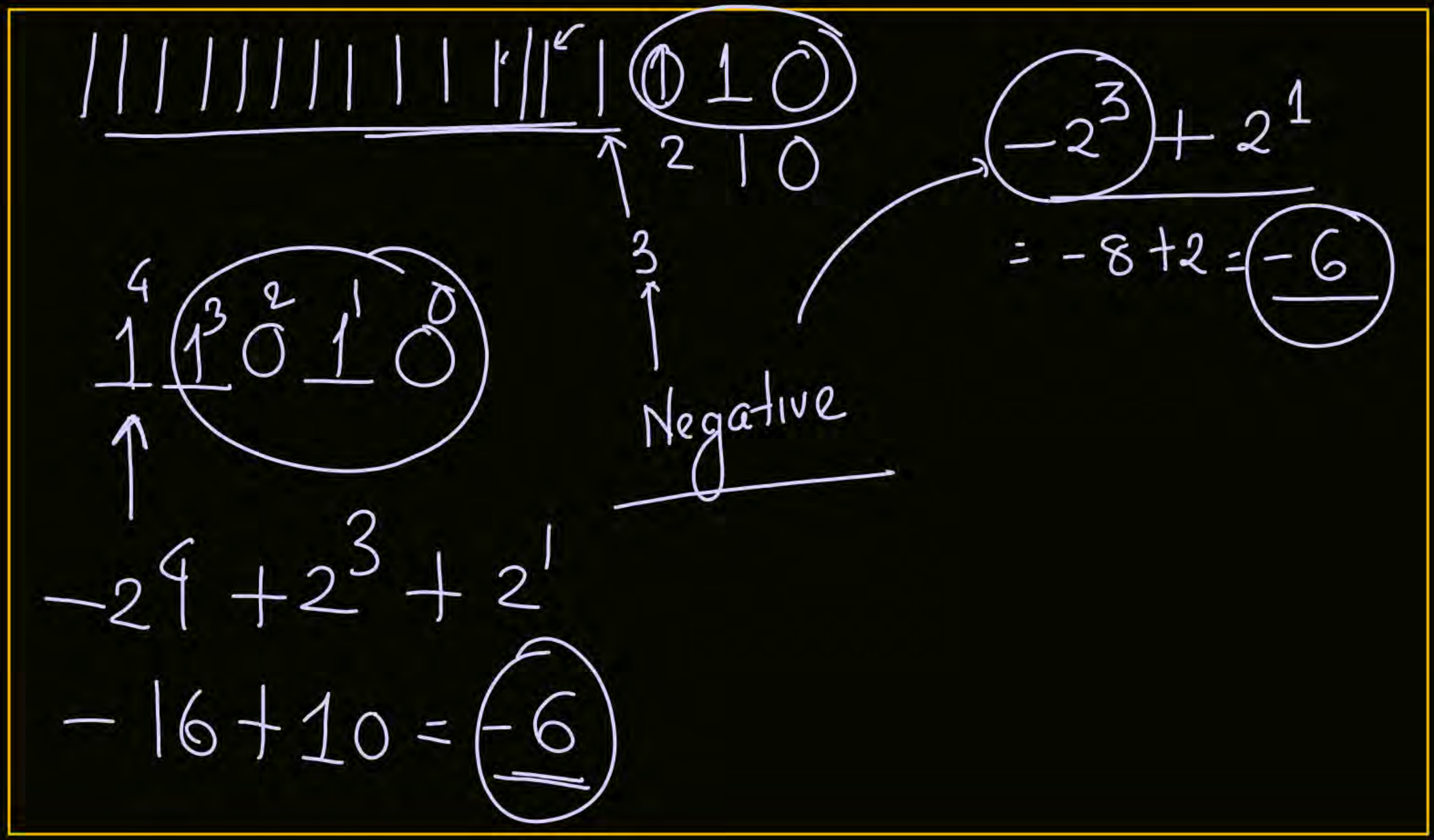
00000101

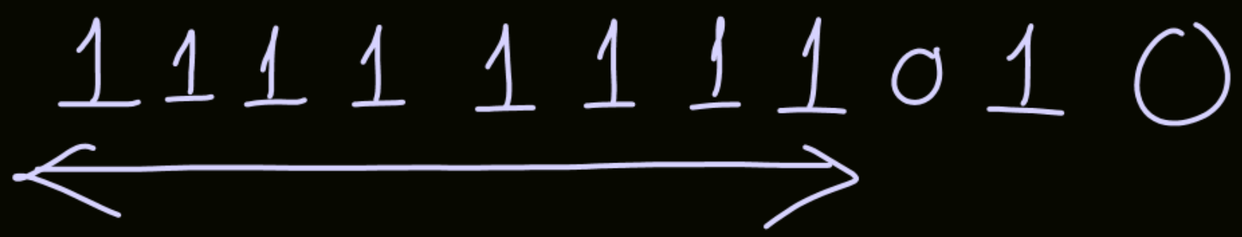
1 1

00000110 (+6)



t.me/Abhishekshamma.pw


1 0 1 0
2 1 0
3
Negative
$-2^3 + 2^1$
$= -8 + 2 = -6$
4
1 1 0 1 0
3 2 1 0
$-2^4 + 2^3 + 2^1$
$-16 + 10 = -6$



Block of 1's

If ignore & reach

last 1 and then take its weight

as Negative and Rest as positive



Bit-wise Operator

8 bits

What is the output the program

```
#include <stdio.h>
    int main ()  {
        int x = 10, z;
        z = ~x;
        printf("%d", z);
        return 0;
}
```

(A) 1

(B) 21

(C) -11

(D) -6

0000 1010

1111 0101

$-2^4+2^2+2^0$

$-2^4+5 = -16+5 = -11$

Slide

Left shift operator

a = 8

a << 2 ← No. of time

1 time shift is equal to multiply by 2

16

2 times shift

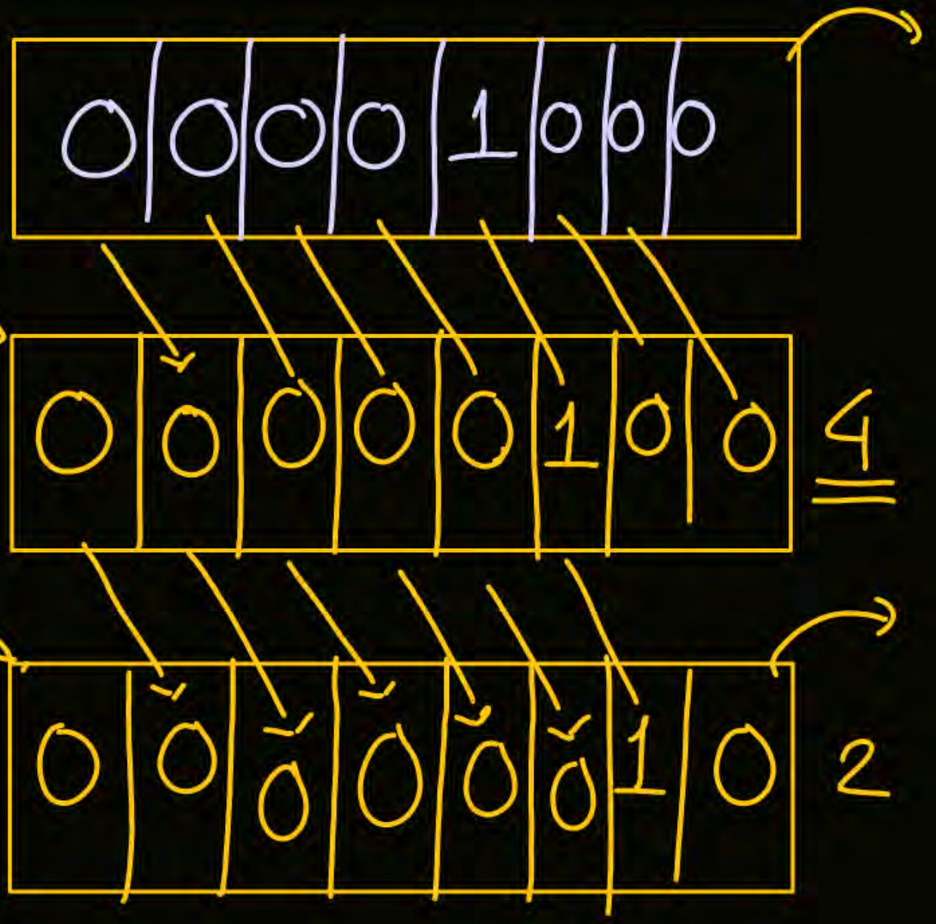
16 × 2 = 82

Right shift

```
int a = 8
a >> 2;
```

1 time shift is equal to divide by 2

2 ~~divide~~ time shift equal to divide by $1/2^2$



```
# include<stdio.h>
int main(){
    int a = -15;
    int b = a>>3;
    printf("%d", b);
    printf("%d",~b);
    return 0;
}
```

H.W



Bit-wise Operator

```
#include<stdio.h>
   int main(){
            char a = 64;
            int k;
            k =a>>3;
            printf("%d", k);
            return 0;
   }
```

(A) 1 (B) 21

(C) 8 (D) -6

Slide



Print() Return value

```
#include<stdio.h>
   int main(){
            char a = 64;
            int k;
            k =a>>3;
            printf("%d", k);
            return 0;
   }
```

(A) 1 (B) 21

(C) 8 (D) -6

Slide



Print() Return value

printf() returns :: No. of character printed

```
int b;
b = printf("Hello Dosto");
printf("%d", b);
```

Hello Dosto11

Slide



Print() Return value

```
#include<stdio.h>
int main(){
   printf("%d",printf("ABCD"));
   return 0; }
```

ABCD4

printf("%d", 4)

(A) ABCD4
(B) 4
(C) ABCD 4
(D) ABCD

Slide



Print() Return value

```
#include <stdio.h>
  int main(){
     printf("\n%d", printf("ABCD"));
     return 0;
   }
```

ABCD

4

/n ← New line character.

Slide



Print() Return value

```
#include <stdio.h>
  int main(){
     printf("\n%d", printf("ABCD"));
     return 0;
   }
```

```
ABCD
4
```



Print() Return value

```
#include <stdio.h>
  int main(){
     int a=1000;
     printf("%d",printf("\n%d",a));
     return 0;
}
```

Output of the program is _____

(A) 1005

(B) 10005

(C) 1000

(D) 1000
    5



Undefined behaviour

```
i++ * ++i; // i is modified more than once
i = ++i    // same as above
++i = 2;   // same as above
i = ++i +1 // same as above
```

Undefined



Ternary Operator

?

execute when expression-true

expression ? expression1 : expression2;

Logical Expression

0 false

1 true

if expression is false then expression 2 execute

Slide



Ternary Operator

```
#include <stdio.h>
int main()
{
    int a;
    a = 10>7?10:20;
    printf("%d", a);
    return 0;
}
```

(A) 10
(B) 20
(C) 1
(D) 0

a = 10>7 ? 10 : 20

true

a = 10

a = 10<7 ? 10 : 20;

Logical expression

false

false a = 20

Slide



Ternary Operator

```
#include<stdio.h>
int main(){
      int x=3, y=4, z=4;
      printf("%d", (z>=y>=x?100:200));
      return 0 ;
}
(a) 100    (b) 200    (c) 0(d) 1
```

[b]

>=

(4>=4)>=3

4>=4>=3

1>=3

1>=3 = F

Left Associative



Control Flow Statement

Sequential Flow of Execution:

Instruction are executed in Sequential manner one after another.

```
int main(){
    S1;
    S2;
    S3;
    S4;
    .
    .
    .
    Sn;
}
```

Slide

Control flow statements can change Sequential flow of execution.

1. Selection or conditional or branching
2. Iterative
3. Jump



Types of Control Flow statement

- Selection or Branching or Decision or Conditional
- Iterative statement
- Jump statements



Selection or Branching or Decision or Conditional

(1) if

(2) if else

(3) if else if

(4) Switch

Nested if



if statement syntax

```
if ( expression ) {

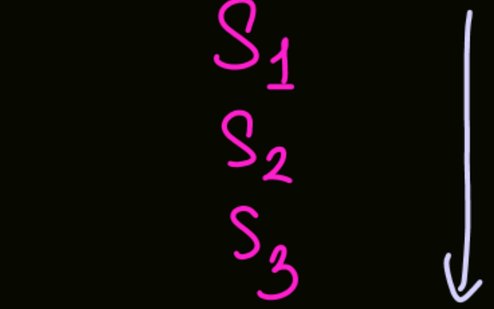
}
```

expression :

- arithmetic
- logical
- Relational

Slide

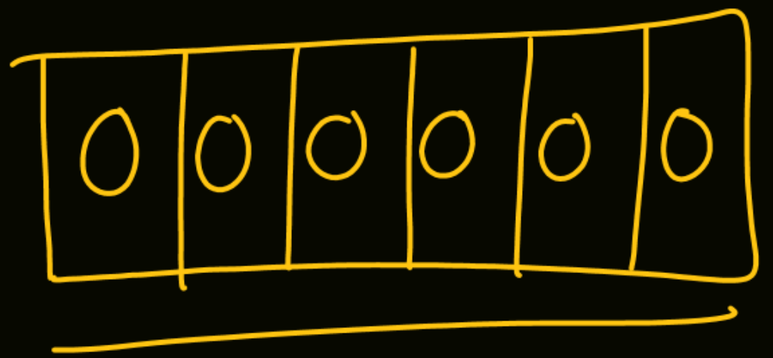
```
int main() {
    S1
    S2
    S3
    if (expression) {
     S4;
     S5;
    }
    S6
    return 0;
    }
```



will only execute if expression evaluate to true

In c language any Non Zero value is true

Zero then false.





f statement flow of execution

```
int a = 10;
if (a >= 10)
    a = a + 10;
printf("%d", a);
```

if(1)

=20

20 value will be printed

```
int a = 5;
if (a >= 10){
    a = a + 10;
}
printf("%d", a);
```

printf = 5

Slide

only Single Stake after if

```
int a = 10;
if (a >= 10)
    a = a + 10;
a = a + 5;
printf("%d", a);
```

Indentation

Not part of if block

if block is Not given

25

```
int a = 5;
if (a >= 10)        → false
    a = a + 10;
a = a + 5;          ← execute
printf("%d", a);
```

Not part of if block.

10

```
int a=5;
if (a=5) {
    a=a+10;
    a=a+5;
}
printf("%d",a);
```

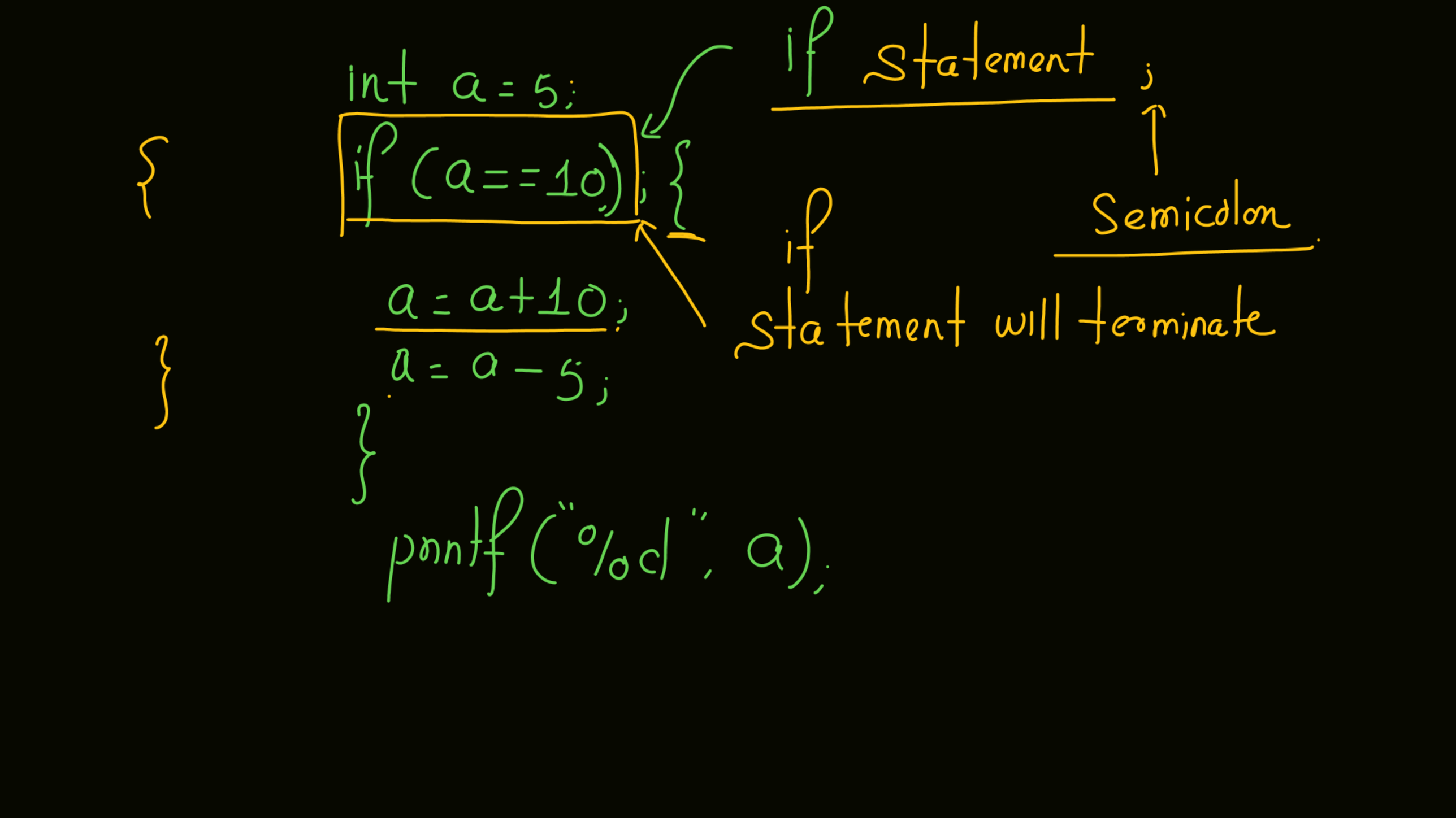
a = 5 → Non Zero

a = 5 → 15

15 + 5 = 20

Because value of a is Non zero if statement is true.

20


if statement ;
Semicolon
if statement will terminate
int a = 5;
if (a==10);
{
a = a+10;
a = a - 5;
}
}
printf("%d", a);

```
int a=5
{
 a=a+10;
 a=a-5;
}
printf("%d", a);
```

~ bitwise complement

~x

expression

Question | ~x = x + 1

In Single expression if a variable modified twice

i = i++; Sequence point.



if statement Example

```
#include <stdio.h>
int main (){
      int a = 10
      if( a < 20) {
            printf("a is less than 20\n" );
      }
      printf("value of a is : %d\n", a);
      return 0;
}
```



if statement Example

```
#include <stdio.h>
      int main (){
            int a = 10
            if(a+2) {
                  printf("Namaskar" );
            }
            printf("Dosto");
            return 0;
}
```



if statement Example

```
#include <stdio.h>
      int main (){
            int a = 10
            if(a-10) {
                  printf("Namaskar" );
            }
            printf("Dosto");
            return 0;
}
```



2 mins Summary

&pic

&pic

&pic

&pic

&pic

Slide

THANK - YOU

